

HUZAIF KHAN

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Full-stack engineer with 4+ years across industry and research and 6+ published papers in edge AI and ML.

SKILLS

Languages: Python, Java, C#, SQL, C, JavaScript, TypeScript
AI/ML: PyTorch, LLMs, RAG, NLP, Transformers, Reinforcement Learning, BioBERT
Frameworks: React.js, Next.js, Node.js, FastAPI, Express.js, Unity, .NET, Xamarin
Cloud & DevOps: AWS, Azure SQL, Docker, GitHub Actions, CI/CD, Git, Linux
Data & APIs: PostgreSQL, MySQL, DataHub, REST APIs, WebSockets
Security & Privacy: Homomorphic Encryption (CKKS), Federated Learning, JWT

WORK EXPERIENCE

CALIFORNIA STATE UNIVERSITY, DOMINGUEZ HILLS

Los Angeles, CA

Graduate Research Assistant - VR/XR with AI & ML

Jan 2026 - Present

- Scaled Unity VR classroom from 5 to 50 concurrent Meta Quest headsets and increased frame rate from 30 to 72 FPS by implementing grid-based culling on standalone hardware.
- Consolidated networked objects from 1,000+ to ~38 and cut per-client bandwidth by 88% by replacing NetworkObject replication with an aggregated Unity Netcode/Relay sync layer.
- Accelerated time-to-first-response from 3.6s to ~350ms by re-streaming a multi-agent Ollama pipeline through FastAPI SSE for responsive voice-driven AI instruction in live VR classroom sessions.

Graduate Research Assistant - Security, Edge AI & ML

Mar 2024 - Jan 2026

- Shortened convergence by 33% (60 to 40 rounds at 0.94 accuracy) across 20-60 clients by offloading CKKS encryption to edge routers and homomorphically aggregating updates.
- Lowered homomorphic-encryption federated learning latency by 46% and encryption overhead by 38% with <1.2% accuracy loss by SIMD-packing CKKS ciphertexts and aggregating them hop-by-hop in-network.

CAPGEMINI TECHNOLOGY SERVICES

Bangalore, India

Senior Software Analyst

Jul 2021 - Jul 2023

- Led the rebuild of a cross-platform Xamarin (C#) time-tracking app for an enterprise client with 3,000+ daily users, replacing separate Android and iOS flows with a shared MVVM navigation layer.
- Reduced clock-in response time by 91% (2s to 180ms) via in-memory caching, async service processing, batched database writes, and fewer client retries.
- Improved average API response time by 64% under peak concurrent load by adding composite indexes to the hot read paths and paginating list views so full result sets weren't loaded per request.

Full-Stack Developer Intern

Jan 2021 - Apr 2021

- Cut page load from 6s to 0.8s as the catalog grew from 10 to 600 products by lazy-loading below-the-fold images, code-splitting the bundle, and deferring non-critical scripts.
- Eliminated duplicate inventory allocations across 100+ concurrent checkout requests by persisting cart state server-side, revalidating stock at checkout, and locking inventory rows during purchase transactions.

PROJECTS

DataHub Steward: Multi-Agent Data Lineage Intelligence

2026

- Trimmed repeat lineage-analysis tokens by 51% (571 to 281) and entities inspected by 40% (30 to 18) by persisting agent findings to DataHub and reusing them across runs.

SenseCLLM: Edge-Resident PII Detection

2026

- Sped up on-device PII detection 10.5x (1960ms to 187ms) and beat Presidio by 49% on F1 by caching LLM-generated regex patterns to a 0.78 hit rate. Published at IEEE Allot 2026.

Medical Text Classification

2025

- Achieved 94.5% accuracy and 0.943 macro F1 across 32 cancer types on 1,905 held-out reports using TF-IDF/Linear SVM with clinical-term masking; extended to BioBERT for contextual representations.

EDUCATION

CALIFORNIA STATE UNIVERSITY, DOMINGUEZ HILLS

Los Angeles, CA

Master of Science, Computer Science - GPA: 3.95/4.0

Aug 2023 - May 2025